

Mumbai MuleSoft Meetup (In Collaboration with Guwahati Meetup Group)

MuleSoft Training for Salesforce Developers and
Beginners - Module 9



Date: 13th Dec Nov 2025
Time: 11 AM to 1 PM



Safe Harbour Statement

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We Love Feedbacks!!! Its Bread & Butter for Meetup.



Organizers/Speakers



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What will we cover in Training?



Module	Topic	What will we cover?	Date
Module 1	Integration & REST/HTTP Basics for Beginners & Salesforce	Integration, P2P, REST APIs, MuleSoft, Anypoint Platform	1 st Nov 2025
Module 2	API Design with RAML for Beginners & Salesforce Developers	API Design with RAML, Publishing APIs to Exchange, Resources	2 nd Nov 2025
Module 3	Anypoint Studio & Mule Basics for Beginners & Salesforce	API Implementation and Deploying API to CloudHub	8 th Nov 2025
Module 4	Core Components, DataWeave & Error Handling Essentials	Dataweave, Error Handling, Core Components	9 th Nov 2025
Module 5	Flow Control & Batch Processing for Scalable Integrations	Batch Processing, For Each, Parallel For Each	15 th Nov 2025
Module 6	HTTP Connector – Listener, Requestor & Payload Handling	HTTP Connector, OAuth Module	16 th Nov 2025
Module 7	Database Connector for CRUD Operations	Database connector to perform query and call store procedure	22 nd Nov 2025

What will we cover in Training?



Module	Topic	What will we cover?	Date
Module 8	Salesforce Connector for Seamless CRM Integration	Deep Dive into Salesforce Connector	23 rd Nov 2025
Module 9	Hosting Options & Deploying Applications to CloudHub	ClodHub 1.0 and CloudHub 2.0	6 th Dec 2025
Module 10	Managing & Securing APIs with API Manager & API Gateway	API Security, API Policies	7 th Dec 2025
Module 11	MuleSoft Demo Project	Database and Salesforce related project	13 th Dec 2025

What have we learned on Day 1?

- What is Point-To-Point Integration?
- What is Integration?
- What is REST APIs?
- What is MuleSoft and Anypoint Platform?
- Walkthrough of Anypoint Platform.
- Understanding the API Lifecycle Management.
- Design the RAML to create and fetch Account and Contacts from Salesforce.
- Published API to Anypoint Exchange.



What have we learned on Day 2?

- What is RAML?
- Reusability of RAML using Traits, Library, Security Schemes.
- OAS (Open API Specification)
- API Governance
- Overview of Anypoint Studio
- Start with API Implementation.



What have we learned on Day 3?

- What Are Connectors
- Salesforce Connector
- Implementing Salesforce System API
- Implementing Properties and Secure Properties
- Deploying MuleSoft Application to CloudHub



What have we learned on Day 4?

- Understanding API Manager and API Gateway
- Creating API Proxy
- Implementing API Auto-Discovery
- Applying Basic Authentication Policy
- Applying Client ID Enforcement Policy



What have we learned on Day 5?

- MuleSoft Batch Processing
- For Each
- Parallel For Each
- Implementing Process API



What have we learned on Day 6?

- MuleSoft Batch Processing
- Error Handling in Batch Processing
- Scatter Gather
- Orchestrating Process API



What have we learned on Day 7?

- Continue with Error handling.
- Building Experience API
- Global Error Handler
- Deploying Process API and Experience API to CloudHub



What have we learned on Day 8?

- Introduction to Dataweave
- Transform Message
- Dataweave Playground
- JSON to XML and XML to JSON
- Dataweave Arrays
- Map and Filter
- MapObject and FilterObject
- GroupBy, Pluck, Reduce
- Calling Java function using Dataweave
- External Dataweave



What will we learn on Day 9?

- Object Store
- Object Store V1 V/S Object Store V2
- Object Store Connector
- Watermarking using Object Store
- Until Successful component



What is Object Store?



In MuleSoft, an Object Store is a persistent or in-memory key–value storage mechanism used by Mule applications to temporarily or permanently store data during runtime. Think of it as a lightweight database or cache that Mule flows can read from and write to.

What is an Object Store?

An Object Store stores objects as key–value pairs, where:

- Key → a unique identifier (usually a string)
- Value → any serializable object (payloads, tokens, variables, etc.)

It is commonly used to maintain state across messages, flows, or even application restarts.



Types of Object Stores

1. In-Memory Object Store

- Data is stored in RAM
- Fast access
- Data is lost when the app restarts
- Best for short-lived or temporary data

2. Persistent Object Store

- Data is stored on disk
- Survives application restarts
- Slightly slower than in-memory
- Best for long-lived state



Object Store Connector



Mule provides an Object Store Connector with operations like:

- Store
- Retrieve
- Remove
- Contains
- Clear

Important to know:

- It's not a database
- Used for temporary or lightweight storage
- Can be in-memory (fast, cleared on restart) or persistent (survives restarts)



Object Store Features



- Supports both static and rolling TTL (time to live) for data persistence.
- Supports using the Connected Apps feature to authenticate with Anypoint Platform using APIs.
- Allows for an unlimited number of entries. There is no limit on the total size of v2 object-stores.
- Stores values up to 10 MB (when Base64 encoded) in size.
- Is available in all supported regions and availability zones within each region.
- Is co-located in the same region as your workers. For example, workers hosted in Singapore would use Object Store v2 hosted in Singapore.
- Provides a Mule connector and REST interface for access by external applications.
- Provides end-to-end secure TLS-transport.
- Encrypts persistent storage to FIPS 140-2 compliant standards.
- Object Store v2 enforces a maximum data persistence of 30 days from time of creation.
- Premium add-on customers are allowed up to 100 TPS per app.
- The free Object Store v2 limits usage to 10 transactions per second (TPS) per application. To upgrade to premium and increase your TPS limit, contact the MuleSoft account team.



Feature	Object Store v1	Object Store v2
Architecture	Legacy, single-region	Cloud-native, distributed
Availability	Limited high availability	Built-in high availability
Scalability	Limited	Auto-scales
Data Persistence	Less reliable	More durable and resilient
Performance	Slower for large workloads	Better performance
TTL (Time to Live)	Supported	Supported (more reliable)
Scope	App-level	App-level and org-level
Regional Support	Single region	Multi-region aware
Management	Minimal controls	Better monitoring & management
Recommendation	Deprecated / legacy	Recommended by MuleSoft



Until Successful



“Until Successful” is a MuleSoft scope that repeatedly executes a set of actions until they complete successfully or until a defined maximum number of retries is reached. It is typically used for operations that may fail temporarily, such as calling external APIs, connecting to databases, or writing to queues.

It helps make Mule applications more resilient by automatically handling transient failures without manual intervention.

Key Features

Feature	Description
Retries on Failure	Retries the enclosed processing until it succeeds.
Max Attempts	Can define a maximum number of retries to prevent infinite loops.
Retry Delay	Can configure the time interval (fixed or exponential) between retries.
Exception Handling	Can handle errors for each attempt and log or route failures.
Persistent Tracking	Works with Object Store (v2 recommended) to track retry attempts across app restarts.

When to Use

- Calling unstable or intermittent APIs
- Writing to external systems with temporary locks or downtime
- Sending messages to busy queues



Until Successful



Until Successful with Failure Validator is a MuleSoft pattern where:

- Until Successful Scope keeps retrying an operation until it passes.
- Failure Validator is used inside the scope to determine whether a failure should trigger a retry or be ignored.
- This allows more granular control over what counts as a retryable failure. Not all errors may need a retry; the failure validator helps decide that.

